Can Skill Substitute for Capital?*

Behavioral Distortions and Stack Asymmetry in Heads-Up Limit Texas Hold'em

Vedant Mundhra

Advised by Professor Jim Dana

April 21st, 2026

*I would like to thank Professor Jim Dana for his continuous guidance and support throughout my semester writing this paper. His elegant ideas at key roadblocks proved invaluable. I am grateful to Professor Shariq Mohammed for an enjoyable and rigorous semester of graduate game theory, which provided the theoretical foundation for this work. Finally, I would like to thank Bobby Squires for many stimulating conversations, late nights, and shared curiosity over mathematics and economics over the course of our undergraduate studies together. All errors are my own.

Abstract

This paper uses computational game theory to analyze how capital constraints and behavioral distortions impact equilibria and outcomes in Texas Hold'em poker. While prior work has established that this game is essentially solved under symmetric starting conditions, no systematic analysis exists of how stack asymmetry affects Nash Equilibrium strategies or how behavioral archetypes perform across varying stack depths. We employ a common algorithm to compute Nash Equilibrium strategies across three analyses. The first varies the stack ratio between players, the second models two behavioral distorted archetypes, and the third synthesizes the two, asking how much of a capital disadvantage a rational player can recover by facing a behaviorally distorted opponent. Stack asymmetry systematically affects equilibrium outcomes, with effects that grow as the short stack becomes more constrained relative to the maximum commitment. Both behavioral archetypes (the maniac and the calling station) are exploitable across all stack depths, with Nash opponents earning significantly above-baseline returns against distorted opponents. The degree of exploitation varies with stack depth, suggesting that capital constraints and preference distortions interact in ways that are not uniform across asymmetry ratios. These findings establish a foundation for understanding how resource inequality and bounded rationality jointly determine competitive outcomes in strategic environments. This paper contributes to literature that largely assumes symmetric, rational players, offering an explanation of how capital constraints and behavioral distortions interact in competitive zero-sum settings.

Contents

1 Introduction **3**

2 Literature Review **3**

3 Methodology and Computational Implementation **5**

3.1 Counterfactual Regret Minimization 5

3.2 Game Abstractions 7

3.3 The Blinds and the Betting Structure 9

3.4 Nash EV and Exploitability 10

3.5 Software Implementation 10

4 Experimental Design **11**

4.1 Research Design 11

4.2 Identification 12

4.3 Behavioral Economics 12

4.4 Utility Parameter Calibration 13

4.5 Predictions 15

5 Results & Discussion **16**

5.1 Overview 16

5.2 Analysis 1: Stack Asymmetry 16

5.3 Analysis 2: Non-Standard Utility 17

5.4 Analysis 3: Combined Effects 17

5.5 Robustness 18

6 Conclusion **19**

A Uniqueness of the Nash Equilibrium Payoff in Two-Player Zero-Sum Games **20**

B Glossary of Terms **22**

1 Introduction

Poker is one of the few casino games in which a player can stand to win money. Since one plays against other players and not the house, the only advantage over a long enough period of time is skill. In 2024, 10,112 people entered the World Series of Poker tournament to compete for a prize pool of $96 million and a 1st place prize of $10 million. The natural question to ask as a player is what the optimal strategy is for any given hand and history. The answer to this lies at the nexus of game theory and computer science.

This paper aims to address a critical gap in literature by analyzing decision-making when different types of players face resource asymmetry. Existing research focuses exclusively on symmetric games where both players begin with equal resources and try to maximize their chips, ignoring the reality that participants rarely enter strategic interactions on equal footing. Our research question is: which suboptimal strategies are the most exploitable at different asymmetric stack depths? This question has implications for understanding competitive dynamics in markets, negotiations, and strategic interactions where participants have heterogeneous budget constraints and varying skill levels.

To answer this question, we employ computational game theory to solve heads-up limit Texas Hold'em under various experimental conditions to find the Nash Equilibrium and measure how equilibrium outcomes change. We proceed with three analyses. The first varies the stack ratio between players, isolating the effect of bankroll asymmetry on equilibrium outcomes. The second varies how rationally one player pursues their objective, isolating the cost of behavioral distortion. The third synthesizes the two, asking how much of a capital disadvantage a skilled player can overcome by facing a less rational opponent, and at what point the stack advantage becomes too large to offset.

This paper extends Nash Equilibrium analysis to settings with heterogeneous budget constraints – an understudied area despite its prevalence in real-world strategic interactions. While behavioral economics has documented decision-making biases extensively, less work has measured their economic cost in competitive zero-sum settings. The insights generalize beyond poker to any environment where participants compete with unequal resources, skill levels, and imperfect information.

2 Literature Review

Zinkevich et al. (2007) introduced Counterfactual Regret Minimization (CFR, described in detail in 3.1), a foundational algorithm for computing approximate Nash Equilibria in large extensive-form games with imperfect information. The key innovation is the concept of "counterfactual regret," which measures how much a player regrets not having chosen a different action at each decision point, weighted by the

probability of reaching that point had they played to reach it. By iteratively minimizing these regrets across all information sets, CFR provably converges to a Nash Equilibrium. The authors demonstrated CFR's effectiveness on poker, solving game abstractions with up to 10^{12} states—approximately twice as large as previous methods. Critically, they also introduced betting abstractions that restricted the number of bets per round to three, reducing computational complexity while maintaining strategic depth. This work established CFR as the dominant approach for solving large imperfect information games and laid the foundation for all subsequent poker AI breakthroughs.

Gilpin and Sandholm (2007) introduced the concept of "lossless abstraction" for extensive-form games with imperfect information, providing a theoretical framework for reducing game complexity without losing strategic information. A lossless abstraction merges game states that are strategically identical thereby reducing the game tree size while preserving the ability to compute exact Nash Equilibria. The authors proved that their abstraction techniques maintain game-theoretic properties: strategies computed in the abstract game can be directly mapped back to optimal strategies in the original game without any loss in solution quality. They applied these techniques to Texas Hold'em poker, creating abstractions based on hand strength and potential, and demonstrated that lossless abstractions could reduce computational requirements by orders of magnitude.

Gibson (2012) introduced Pure CFR, a Monte Carlo variant of CFR that samples a single pure strategy profile per iteration rather than traversing the full game tree. This reduces memory requirements substantially at the cost of higher variance per iteration, with the same theoretical convergence guarantee as vanilla CFR. The key insight is that by sampling a single pure strategy (one that assigns probability 1 to a single action at every information set) the algorithm avoids storing and updating values across the entire game tree on each pass, making it tractable for large games with many information sets. Pure CFR forms the computational backbone of this analysis; we adopt it over CFR+ specifically for its bucket-level memory storage, which reduces peak memory requirements from approximately 3TB to under 500GB for our 100-bucket abstraction.

Bowling et al. (2015) demonstrated that heads-up limit Texas Hold'em poker has been "essentially weakly solved" using an improved variant of Counterfactual Regret Minimization called CFR+. Their program, named Cepheus, computed a strategy so close to optimal that even if a perfect opponent played against it for a human lifetime (assuming one hand per minute for 70 years), the opponent would be statistically guaranteed not to win money—losing at most 0.001 big blinds per game on average. This breakthrough required solving a game with 3.19×10^{14} decision points and took approximately 900 CPU-core years of computation, resulting in a 262-terabyte strategy table. The achievement marks the first time a non-trivial imperfect information game has been essentially solved, demonstrating that game-theoretic optimal strategies

can be computed for games larger than any previously solved while accounting for hidden information.

3 Methodology and Computational Implementation

The methodology exploits the fact that poker is a zero sum game. Zero sum games have have a unique Nash equilibrium payoff for each player. This implies that even in complex extensive form games, the payoff in any subgame perfect Nash equilibrium is the same as the payoff in any Nash equilibrium (since all SPNE are Nash equilibrium). So finding a Nash equilibrium is almost certainly very close to finding a subgame perfect equilibrium.

3.1 Counterfactual Regret Minimization

Counterfactual Regret Minimization (CFR) is the predominant algorithmic framework for computing approximate Nash Equilibria in large extensive-form games with imperfect information. The algorithm has underpinned major breakthroughs in poker AI, including the solution of heads-up limit Texas Hold'em (Bowling et al., 2015), the defeat of professional players in no-limit poker (Brown & Sandholm, 2017), and the development of superhuman multiplayer agents (Brown & Sandholm, 2019).

The core idea behind CFR is straightforward: the algorithm learns optimal strategies by tracking regret for actions not taken. At each decision point in the game, the algorithm calculates how much better or worse each available action would have been compared to the player's actual mixed strategy. For player i when choosing a mixed strategy at information set I, the instantaneous regret for action a with probability 1 at iteration t of the algorithm is

$$r_i^t(I, a) = v_i^t(I, a) - v_i^t(I)$$

where $v^t(I, a)$ is the expected payoff of always taking action a at information set I and following the current strategy thereafter, and $v^t(I)$ is the expected payoff of the current mixed strategy at I.

$$v_i^t(I) = \sum_{a' \in A(I)} \sigma_i^t(I, a')\, v_i^t(I, a')$$

where $\sigma_i^t(I, a')$ is the probability of action a' at information set I and $A(I)$ is the set of feasible actions at information set I. The cumulative regret over all iterations is

$$R^t(I, a) = \sum_{\tau=1}^{t} r^\tau(I, a)$$

Actions that would have performed better accumulate positive regret, while actions that would have per-

formed worse accumulate negative regret. We ignore negative regret because it simply means that switching to that action would not have improved our payoff, so only positive regret provides a meaningful signal for adjusting the strategy. Over many iterations, the algorithm adjusts its strategy to play actions proportionally to their positive regrets, which shift probabilities toward actions the player "regrets" not having chosen more often. This allows the algorithm to converge to a mixed strategy; it does not assign a probability of 1 to the highest payoff choice.

The "counterfactual" component refers to how regrets are calculated. Rather than only learning from game paths actually taken, CFR considers hypothetical scenarios where the player plays to reach specific decision points, even if their current strategy would rarely lead there. At each information set, regrets are weighted by the probability that the opponent's strategy would have led to that decision point, isolating the player's own strategic choices from their opponent's actions. This counterfactual reasoning allows the algorithm to explore the full strategic space efficiently. The algorithm is optimizing even at information sets which might never be reached, thereby solving for a subgame perfect equilibrium.

CFR updates strategies through regret matching. At each information set, the probability of taking each possible action becomes proportional to that action's cumulative positive regret. Let $\sigma^t(I)$ denote the strategy (probability distribution over actions) at information set I during iteration t. For each action $a \in A(I)$ where $A(I)$ is the set of available actions at information set I, period t is used to generate the strategy at iteration $t+1$ as

$$\sigma^{t+1}(I,a) = \frac{\max(0, R^t(I,a))}{\sum_{b \in A(I)} \max(0, R^t(I,b))}$$

where $\sigma^{t+1}(I,a)$ represents the probability of selecting action a at information set I, with $\sum_{a \in A(I)} \sigma^{t+1}(I,a) = 1$. Actions with negative cumulative regret receive zero probability in the numerator, while actions with positive regret are weighted proportionally. If all regrets are non-positive (denominator equals zero), a uniform random strategy is used.

The key theoretical result is that if all players minimize their counterfactual regret, the average strategy profile converges to a Nash Equilibrium. Zinkevich et al. (2007) prove that in two-player zero-sum games, if each player's average regret diminishes with the number of iterations, this average strategy profile is an approximate Nash Equilibrium. Specifically, if player i's average regret after T iterations is at most ϵ, then the strategy profile is a 2ϵ-Nash Equilibrium. The Nash Equilibrium approximation is the average strategy across all iterations, not the final iteration's strategy. The average strategy over T iterations is defined as:

$$\bar{\sigma}^T(I,a) = \frac{1}{T} \sum_{t=1}^{T} \sigma^t(I,a)$$

Individual iteration strategies fluctuate as the algorithm explores, but their average has been proven to converge to equilibrium. Because this is a two-player zero-sum game, the Nash Equilibrium payoff is unique regardless of which equilibrium strategy profile is selected, eliminating concerns about equilibrium selection, subgame perfection, or perfect Bayesian refinements (see Appendix A).

We employ Pure CFR, a variant of Counterfactual Regret Minimization introduced by Gibson (2012). Rather than traversing the full game tree on each iteration, Pure CFR samples a single pure strategy profile per iteration, reducing memory requirements at the cost of higher variance per iteration. The convergence guarantee is the same as vanilla CFR making Pure CFR theoretically equivalent to its variants for the purposes of this analysis. While Bowling et al. (2015) employed CFR+ to solve the symmetric heads-up limit hold'em game, we adopt Pure CFR for its bucket-level memory storage, which reduces peak memory requirements from approximately 3TB to under 500GB for our 100-bucket abstraction. This is a deliberate engineering decision rather than a theoretical compromise. The two algorithms belong to the same family with identical convergence guarantees, and the memory reduction is critical given the scale of the asymmetric stack experiments conducted here.

The choice of Pure CFR is further motivated by its scalability to large games through recursive tree traversal, natural handling of imperfect information through information set-based calculations, and the validated implementation provided by the open-pure-cfr-buckets solver, which forms the computational backbone of this analysis.

3.2 Game Abstractions

Solving heads-up limit Texas Hold'em directly is computationally infeasible due to the game's enormous state space. There exist greater than 10^{13} information sets even in the limit variant. To make the problem manageable, we employ three strategic abstractions that reduce game complexity while preserving essential strategic structure. Following Gilpin and Sandholm (2007), we distinguish between lossless abstractions that introduce no approximation error and lossy abstractions that trade accuracy for computational feasibility.

3.2.1 Suit Isomorphism

Suit isomorphism is a lossless abstraction that exploits the symmetry of card suits in poker. Before any community cards are revealed, hands differing only in suit assignment are strategically identical. For example, holding $A\spadesuit K\spadesuit$ is equivalent to holding $A\heartsuit K\heartsuit$ since no suit has an inherent advantage. Formally, two hands are suit-isomorphic if one can be transformed into the other through a permutation of suits that preserves all strategic properties. This abstraction reduces the preflop hand space from 1,326 distinct

two-card combinations to 169 strategically unique hand types.

Suit isomorphism is lossless because it merges only truly equivalent states. The Nash Equilibrium strategy computed in the reduced game maps perfectly to the optimal strategy in the full game. For example, the hands $\{A\spadesuit K\spadesuit,\ A\heartsuit K\heartsuit,\ A\diamondsuit K\diamondsuit,\ A\clubsuit K\clubsuit\}$ all map to the exact same preflop information set, reducing the game tree by a factor of approximately eight with zero information loss.

3.2.2 Card Bucketing

Card bucketing is a lossy abstraction that groups strategically similar hands into clusters to further reduce the state space. Card bucketing groups strategically similar hand-board combinations by imposing that strategies must be identical across all information sets within the same bucket. Without bucketing, each information set is defined by the player's private cards and the complete betting history; on the river, this creates over two million distinct information sets per player. Bucketing merges these information sets: all hands assigned to the same bucket use a single shared strategy regardless of their specific card combinations.

We employ 100 buckets per betting round (flop, turn, river) generated via k-means clustering using the Opponent Cluster Hand Strength (OCHS) metric, implemented in the poker-cfrm library. Rather than summarizing hand strength as a single number, OCHS measures how a hand performs against each cluster of opponent holdings, producing one vector per opponent per bucket of win probabilities that captures the full strategic profile of a hand across the range of possible opponent hands. Two hands are considered similar not because they have similar raw strength, but because they win and lose against the same types of opponent hands in similar proportions. Earth Mover's Distance is used to compare these vectors during clustering, grouping hands with similar performance profiles into the same bucket. This produces a richer and more strategically meaningful abstraction than simpler metrics such as Expected Hand Strength (EHS), which only measures the probability of winning against a random opponent hand. EHS is computed and stored as a lookup table used internally by the clustering pipeline, but the final bucketing metric is OCHS. Preflop, suit isomorphism alone reduces the hand space to 169 canonical hand types and no further bucketing is needed.

Card bucketing is not built into the open-pure-cfr-buckets solver and must be generated externally using the poker-cfrm library, which implements k-means clustering using the Opponent Cluster Hand Strength (OCHS) metric. The generated bucketing scheme is saved to a file and loaded by the solver during CFR computation. For example, on a flop of $7\heartsuit 4\diamondsuit 2\clubsuit$, hands like $\{A\spadesuit K\heartsuit,\ A\clubsuit Q\diamondsuit\}$ (high cards with no pair) might all map to the same bucket, while $\{A\spadesuit A\clubsuit,\ Q\heartsuit Q\diamondsuit\}$ (overpairs) map to a different bucket.

3.2.3 Betting Cap

The betting cap is a lossy abstraction that limits the maximum number of raises allowed per betting round. In standard limit Texas Hold'em, players may raise up to four times per round. We impose a three-bet cap: after three raises in a round (2 raises for the preflop round after the big blind has been posted), the next player may only call or fold. This constraint is similar to the approach used by Zinkevich et al. (2007). This abstraction is lossy because it eliminates legitimate strategic sequences where four or more raises might occur. However, empirical analysis of poker hand histories suggests that betting sequences reaching four raises occur in less than one percent of hands. The betting cap substantially reduces game tree depth while affecting an extremely small fraction of strategic situations.

3.3 The Blinds and the Betting Structure

We look at 2 player (or Heads-up) Texas Hold'em Poker with betting limits. Texas Hold'em is played across four betting rounds, or "streets": preflop (after 2 cards dealt to each player), flop (after 3 common cards dealt), turn (after a 4th common card dealt), and river (after a 5th common card dealt). One player is assigned to be the small blind and posts 1 chip to the pot. The other player is assigned to be the big blind and posts 2 chips to the pot. It is convenient to refer to bets in units of the big blind (BB), which is 2 chips. Bets on the preflop and flop are made in increments of 1BB (2 chips), while bets on the turn and river are made in increments of 2BB (4 chips). Raises are capped at 2 chips per bet for preflop and flop and at 4 chips per bet on the turn and river.

The big stack is fixed at 36 chips (18BB) across all experiments. This value is not arbitrary. It is the smallest stack size satisfying three simultaneous constraints. First, the solver requires blind values to be integer numbers of chips, which constrains admissible stack sizes. Second, across the four stack ratios considered (1:1, 1:2, 1:3, 1:4), the short stack must also be an integer number of chips, requiring the big stack to be divisible by 1, 2, 3, and 4. Third, and most importantly, 36 chips exactly equals the maximum theoretical commitment a single player can make in one hand under the 3-bet cap structure:

Street	Calculation	Chips
Preflop	$2 + 2 \times 2$	6
Flop	3×2	6
Turn	3×4	12
River	3×4	12
Total		**36**

Setting the big stack to exactly this value ensures the big stack player is never chip-constrained, regardless of how the betting unfolds, while any larger value would be strategically irrelevant. 36 chips is therefore the tightest possible unconstrained stack, and the three constraints together determine it uniquely.

The short stack varies across experiments: 36 chips (1:1), 18 chips (1:2), 12 chips (1:3), and 9 chips (1:4). As the ratio increases, the short stack covers a decreasing fraction of the maximum possible commitment (100%, 50%, 33%, and 25% respectively) progressively constraining the short stack player's ability to credibly threaten full commitment across all streets.

3.4 Nash EV and Exploitability

The standard metric for evaluating strategies in the poker theory literature is exploitability which is defined as the gap between a strategy's expected value and the Nash Equilibrium expected value against a worst-case best-response opponent (Zinkevich et al., 2007). A fully Nash Equilibrium strategy has exploitability of zero by definition; deviations from equilibrium open the player to targeted punishment.

We use Nash EV rather than formal exploitability. Nash EV is defined here as the average chip gain per hand for the player without any behavioral distortion facing an opponent whose strategy is the CFR equilibrium of a distorted utility game. The two metrics are related but distinct: exploitability asks how badly a strategy performs in the worst case, whereas Nash EV asks how much a Nash opponent profits in expectation from facing a particular behavioral type. The latter makes more sense as the economic question of interest. We are not quantifying the worst-case vulnerability of the distorted utility strategies. We are measuring the equilibrium profit a Nash player extracts from each archetype across stack configurations.

There is also a computational motivation. Formal exploitability requires a full best-response traversal of the game tree on top of the CFR solve, which is expensive and introduces additional programming complexity. Nash EV is measured directly via Monte Carlo simulations: both strategies are fixed post-solve and played out over a large number of hands, with the Nash player's average chip gain recorded as the output. All simulations reported here use 5 million hands to wash out any variance.

3.5 Software Implementation

Our implementation uses two open-source tools that handle different parts of the problem. For generating card abstractions we use the poker-cfrm library (`https://github.com/pandaant/poker-cfrm`), a C++ implementation designed specifically for poker hand clustering. For solving the game itself, we use the open-pure-cfr-buckets solver, which implements Pure CFR with bucket-level memory storage and forms the

computational backbone of all experiments reported here.[1]

The workflow combines these tools as follows. First, we use poker-cfrm to generate our card bucketing scheme offline, producing an abstraction file that assigns bucket labels to hand-board combinations based on OCHS clustering. Second, we load this abstraction into the solver and run Pure CFR on the bucketed game, computing approximate Nash Equilibrium strategies. The entire CFR computation operates on the abstracted game. Once a strategy has converged, Nash Expected Value (Nash EV) is measured via Monte Carlo simulation: both strategies are fixed and played out over 5 million randomly dealt hands, with the average chip gain per hand recorded as the output.

4 Experimental Design

4.1 Research Design

This paper estimates the effect of two distinct sources of strategic distortion on equilibrium outcomes in heads-up limit Texas Hold'em poker. These distortions, described below, are realistic and important. We consider distortions that are widely perceived as common among actual poker players. The goal is to better understand how costly certain common behaviors are for the player.

There are two players in the game. Player 0 posts the small blind and acts first on all post-flop streets, and Player 1 posts the big blind and acts last post-flop. The two occupy structurally asymmetric positions regardless of stack size. The outcome of interest is Nash Equilibrium expected value (Nash EV), defined as the average chip gain per hand for Player 0 when both players play their CFR-derived Nash strategies across a large number of randomly dealt hands. Nash EV is measured in milli-big-blinds per hand (mbb/hand), where 1 big blind equals 2 chips in our game specification.

We proceed across three analyses. The first holds utility functions fixed at the standard chip-maximizing assumption and varies the stack ratio between players. The second holds stacks fixed at the symmetric baseline and varies the utility function governing Player 1's preferences. The third combines the results of the first two to derive exchange rates between skill and capital. Together, the first two analyses isolate the contributions of capital constraints and suboptimal behavioral distortions to equilibrium outcomes, which the third analysis then combines.

[1] All code, solver configurations, and experimental scripts are available at `https://github.com/vedantm7/senior-undergraduate-thesis`.

4.2 Identification

A core identification challenge in all analysis arise from the positional structure of the game. Because Player 0 acts out of position on every post-flop street, assigning the short stack to Player 0 in the experiments would conflate two distinct effects: the disadvantage of holding fewer chips and the disadvantage of acting first without knowledge of the opponent's response. Any observed change in Nash EV relative to the symmetric baseline would then be attributable to either source, or both, with no way to separate them.

To isolate the pure effect of stack asymmetry, we solve and simulate each asymmetric configuration twice—once with the short stack assigned to Player 0 and once with the short stack assigned to Player 1—then average the two estimated effects. Let $\mathrm{EV}_A(r)$ denote the Nash EV for Player 0 under stack ratio r when the short stack is assigned to Player 0, and let $\mathrm{EV}_B(r)$ denote the corresponding value when the short stack is assigned to Player 1. The pure stack effect at ratio r is recovered as:

$$\Delta_{\text{stack}}(r) = \frac{\left[\,(\mathrm{EV}_A(r) - \mathrm{EV}_{\text{base},A}) - (\mathrm{EV}_B(r) - \mathrm{EV}_{\text{base},B})\,\right]}{2}$$

where $\mathrm{EV}_{\text{base},A}$ and $\mathrm{EV}_{\text{base},B}$ are the symmetric baseline EVs from each positional assignment. Because the baseline game is symmetric in stacks, $\mathrm{EV}_{\text{base},A} \approx -\mathrm{EV}_{\text{base},B}$, so the positional component cancels in the difference and the estimator recovers the chip disadvantage attributable to the stack constraint alone. This design doubles the number of required solves but eliminates a confounder that would otherwise be impossible to address after the experiments had been run.

4.3 Behavioral Economics

The three archetypes (which we call the *maniac*, the *calling station*, and the *nit*) studied in Analysis 2 are motivated by a behavioral economics literature on systematic deviations from expected utility maximization. Rather than modeling suboptimal play as random error, we follow the tradition of Kahneman and Tversky (1979) and Simon (1955) by treating deviations as structured. For example, this tradition has looked for various biases in consumers, such as caring more about the present and less about the future (time inconsistent preferences). Players have well-defined preferences — just ones that diverge from pure chip maximization.

A natural approach would be to directly change the strategy at each information set. For example, fixing raise probabilities above their Nash values to simulate a maniac. This approach is inadequate for a subtle reason: CFR would just reoptimize. If we artificially add (or multiply by) x to the raise probability at a given information set, CFR will simply reduce the base raise probability by x during optimization, leaving the net raise frequency unchanged.

Instead, we draw from behavioral economics literature and model each archetype by modifying the utility function itself. This is conceptually important: under this framing, the maniac, calling station, and nit are not suboptimal players at all. Each is an agent maximizing their own well-defined objective. It simply happens to be that their objective differs from chip maximization. CFR then finds the Nash Equilibrium strategy with respect to each distorted utility function, producing behavior that is optimal given the player's preferences.

The three utility functions are defined as follows. The standard Nash player maximizes expected value of chips directly, $U(x) = x$. The maniac receives utility from expected value of chips and a utility bonus of α for each raise taken across all streets, $U = x + \alpha \cdot (\text{raises})$, representing a preference for aggression independent of chip outcome. The calling station receives utility from expected value of chips and sustains a utility penalty of β for each fold taken across all streets, $U = x - \beta \cdot (\text{folds})$, representing discomfort with passive surrender independent of chip outcome. The loss-averse nit is defined by a piecewise linear utility function: $U = x$ when $x \geq 0$, and $U = \lambda \cdot x$ when $x < 0$. Wins are valued at face value, but losses are amplified by factor λ, producing excessive risk aversion and a tendency to over-fold.

This framing draws directly from bounded rationality (Simon, 1955) and the prospect theory framework of Kahneman and Tversky (1979). The maniac and calling station follow Simon's framing naturally: each player is optimizing, just with a different objective that incorporates action-specific preferences. The loss-averse nit uses the pure loss aversion specification with $\lambda = 1.31$ (Walasek, Mullett, and Stewart, 2024). We adopt only the loss aversion component of prospect theory. The utility function is piecewise linear with a kink at zero, without the probability weighting or value function curvature of the full Kahneman-Tversky framework. In all three cases, the player is not 'suboptimal' in the formal sense. They are rational agents whose preferences happen to diverge from pure chip maximization. It is precisely this divergence that a Nash opponent can identify and exploit.

4.4 Utility Parameter Calibration

One of the methodological challenges we ran into for players whose utility functions are not expected chip maximization was the choice of utility parameters. The distorted utility functions each contain a free parameter (α and β) whose value determines how extreme the behavioral distortion is. We initially considered two naive approaches which were both unsatisfying. The first was to calibrate parameters such that each archetype loses a set number of big blinds per hand against a Nash opponent. This inverts the logic of the experiment, mechanically fixing the conclusion before the analysis begins. The second was to pick parameters that "seem right" intuitively, but this leaves the results entirely dependent on an arbitrary

choice, undermining the interpretation of the findings. Instead, the approach we used is described below.

Calibration targets for the maniac and calling station archetypes are derived empirically from the HandHQ dataset, a publicly available collection of 21,605,687 real-money online poker hands. The dataset was filtered to heads-up tables and restricted to regular players with stable behavioral tendencies. Action frequencies were computed per player per session, requiring a minimum of 50 action opportunities per session for stable estimates, and aggregated into empirical distributions of raise frequency and fold frequency across the filtered sample.

It is worth noting that HandHQ contains no-limit hold'em hands, whereas our focus is limit hold'em. The dataset is used strictly as a reference for calibrating archetype extremity, not as a strategic comparison. The question it helps us answer is not "how do limit players behave" but rather "what raise and fold frequencies represent extreme but empirically documented human behavior."

The maniac calibration target of 0.55–0.60 raise frequency corresponds to above the 95th percentile of the observed raise frequency distribution. The calling station calibration target of 0.10–0.12 fold frequency corresponds to below the 10th percentile of the observed fold frequency distribution. Both represent behavior that is extreme relative to the population but documented in real play, grounding the archetypes in observable human tendencies rather than arbitrary parameter choices.

The calling station calibration converged cleanly. A penalty parameter of $\beta = 5.0$ produces an all-streets fold frequency of 0.124, close enough to the target range to be accepted as the final calibrated value. The maniac calibration proved more constrained. At high values of α, the all-streets raise frequency asymptotes around 0.41–0.43 rather than reaching the empirical target of 0.55–0.60. Although we were limited by time for a full diagnosis, we had a theory: the 3-bet cap limits the number of raises available per street, and the Nash opponent adapts to increased aggression by tightening, which itself suppresses the achievable raise frequency. The final calibrated value of $\alpha = 5.0$ produces a raise frequency of approximately 0.41. Rather than matching the empirical P95 exactly, the maniac archetype will be characterized as highly aggressive relative to the Nash baseline.

A loss-averse nit archetype was initially planned using the empirically grounded loss aversion parameter $\lambda = 1.31$ from Walasek, Mullett, and Stewart (2024). However, the subtle utility distortion implied by this parameter value was too small to produce stable, theoretically consistent behavioral effects when implemented in the computational solver. The nit archetype was dropped as a result; this limitation is discussed further in the robustness section.

4.5 Predictions

4.5.1 Analysis 1: Stack Asymmetry

A player with a binding stack constraint has a reduced action space relative to a deep-stacked opponent: they will not be able to credibly threaten large raises or re-raises, their all-in threat carries less value, and they are forced into all-in situations earlier in the betting tree where informational disadvantages compound. This is similar to a firm facing a hard borrowing constraint competing against a financially unconstrained rival. The constrained player cannot make the same credible commitments and is therefore exploitable where the unconstrained player is not. We therefore predict: Nash EV for the short-stacked player decreases monotonically as the stack ratio increases from 1:1 to 1:4, after controlling for positional effects via the symmetric assignment estimator.

4.5.2 Analysis 2: Non-Standard Utility

We predict Nash EV for Player 0 increases monotonically in α. Larger aggression bonuses induce more over-raising by Player 1 in spots where the Nash strategy would call or fold, leaking chip value by building pots in which Player 1 is not the favorite. Nash EV for Player 0 increases monotonically in β. Larger fold penalties induce Player 1 to call down with losing hands, allowing Player 0 to extract value through straightforward value betting in spots where a Nash opponent would have folded.

4.5.3 Analysis 3: Combined Effects

We predict behavioral distortions will partially but not fully offset capital disadvantages, with the degree of recovery declining as the stack ratio increases. At moderate asymmetry, the distortion premium should be large enough to meaningfully close the gap opened by the stack penalty; at extreme asymmetry, the capital advantage should dominate regardless of behavioral type. The combined EV for the short-stacked Nash player should be approximately equal to the sum of the stack penalty from Analysis 1 and the distortion premium from Analysis 2.

5 Results & Discussion

5.1 Overview

Table 1: Nash EV for Player 0 by Opponent Utility Type and Stack Configuration (mbb/hand)

Player 1 Utility Type	36v36	18v36	12v36	9v36
Standard (Nash)	−73.40	−34.21	−33.64	−48.49
vs. Station (β)	+117.99	−1.44	−65.30	−1.99
vs. Maniac (α)	+205.98	+184.08	+44.65	excluded[2]

Table 1 reports difference-in-means estimates of the stack and behavioral distortion effects, measured in milli-big-blinds per hand (mbb/hand), net of positional effects. One milli-big-blind is one thousandth of the big blind, a standard unit for measuring per-hand gains in poker. Player 1's utility type varies across rows and stack configuration varies across columns, where the ratio x:y indicates Player 0 holds x chips and Player 1 holds y chips. The top left cell is the symmetric baseline from which deviations are measured. Three preliminary findings emerge. First, stack asymmetry imposes a negative EV on the short-stacked player relative to the symmetric baseline, with effects that are non-monotonic across stack ratios. Second, both behavioral archetypes are exploitable by a rational opponent with symmetric stacks, with the maniac being more exploitable than the calling station. Third, the interaction between stack asymmetry and behavioral distortion is not uniform. Nash EV varies across stack depths in ways that differ between archetypes suggesting that bankroll asymmetry and behavioral distortions are not independent. A more detailed analysis follows.

5.2 Analysis 1: Stack Asymmetry

The first row in Table 1 reports Player 0's EV as Player 1's stack advantage changes under the standard chip-maximizing utility specification. In the symmetric baseline, Player 0 earns -73.40 mbb/hand. This is not a stack effect but rather the structural cost of being player 0 in the game itself. In heads-up limit hold'em, even perfect play against a perfect opponent does not produce zero EV because the actions of player 0 and player 1 are not perfectly symmetric. The blind structure imposes a persistent per-hand cost on players, and -73.40 mbb/hand represents the equilibrium price of that disadvantage. All subsequent results are interpreted relative to this baseline.

[2]The maniac 9v36 configuration was excluded due to a power outage on Northeastern's computing cluster.

Across all three asymmetric configurations, Player 0's pure stack effect is less negative than the symmetric baseline, suggesting that the short-stacked player systematically benefits from the asymmetry once positional effects are controlled for. A likely mechanism is commitment credibility: a short stack pre-commits Player 0 to a narrow action space, removing the multi-street pressure a deep opponent would otherwise apply and forcing Player 1 to account for the credible all-in threat at every decision point. The capital constraint, paradoxically, simplifies the strategic environment in ways that favor the constrained player.

5.3 Analysis 2: Non-Standard Utility

The 36v36 column reports Player 0's Nash EV gain under symmetric stacks when Player 1 optimizes a distorted utility function. Both behavioral distortions generate substantial positive Nash EV for Player 0 even without any stack asymmetry, confirming that non-standard preferences are exploitable by a rational opponent in isolation. The maniac distortion generates the larger gain (+205.98 mbb/hand), followed by the calling station (+117.99 mbb/hand), suggesting that over-aggression is more exploitable than over-passivity under symmetric stacks.

The maniac's inflated raise frequency creates opportunities for Player 0 to call bets profitably with hands that would not otherwise warrant a call against a Nash opponent. The calling station's reluctance to fold allows Player 0 to extract value through straightforward value betting across a wide range of holdings, though the effect is smaller in magnitude likely because passive opponents deny Player 0 the pot-building opportunities that aggressive opponents inadvertently create.

5.4 Analysis 3: Combined Effects

The asymmetric behavioral cells in Table 1 report Player 0's Nash EV when Player 1 holds both a stack advantage and a distorted utility function. This is the central experiment of the paper: it asks whether a rational but capital-constrained player can recover EV losses through the exploitation of a behaviorally distorted opponent, and at what stack ratios behavioral distortion is sufficient to overcome the capital disadvantage entirely.

The results reveal fascinating patterns that differ between archetypes. Against the maniac, behavioral distortion reverses the short stack penalty entirely at all measured stack depths. Player 0 earns +184.08 and +44.65 mbb/hand at 1:2 and 1:3 respectively, suggesting that over-aggression creates exploitable opportunities large enough to dominate the stack disadvantage.

Against the calling station, the results are more complex. The stack penalty is nearly completely recovered at 1:2 and 1:4 (-1.44 and -1.99 mbb/hand respectively), but at 1:3 the combined effect is -65.30 mbb/hand

which is worse than facing a Nash opponent at the same stack depth. This non-monotonicity suggests that something about the interaction between stack depth and calling station behavior is not uniform, and that certain stack configurations amplify rather than lessen the capital disadvantage against passive opponents.

These findings complicate the simple skill-versus-capital substitution narrative. Against aggressive opponents, a rational player can easily compensate for capital constraints across all depths. Against passive opponents, the relationship is depth-dependent and non-linear, with moderate asymmetry (1:3) as the worst configuration for the short-stacked rational player.

5.5 Robustness

Several features of the computational design support the reliability of these results. The baseline CFR solve ran for 255 billion Pure CFR iterations over approximately five days of wall time, achieving a convergence criterion of 0.075 percent average action probability change between the final two hourly checkpoints. A zero-sum validation confirmed that Player 0 and Player 1 average EVs sum to exactly zero across 5,000,000 simulated hands, verifying the mechanical correctness of the Nash EV simulation. The difference-in-means design directly addresses the positional confound, and all experimental configurations use identical solver settings, abstraction files, and convergence criteria. Across all configurations estimated, EV results were stable at two decimal places across repeated simulations with different random seeds, confirming that Monte Carlo sample sizes were sufficient to eliminate simulation variance as a confound.

There is one notable observation about the magnitude of the behavioral distortions. The maniac archetype does not reach its empirical calibration target. Real-money hand histories suggest maniacs raise at frequencies of 0.55 to 0.60, yet the CFR equilibrium under the maniac utility function asymptotes well below this at 0.41. The calling station, by contrast, achieves a fold frequency of 0.12 that falls at the 10th percentile of observed behavior. Despite the maniac being below its target archetype severity, it generates higher EV for the rational player than the calling station across all stack depths. This suggests that over-aggression is disproportionately costly relative to its magnitude compared to over-passivity — a non-obvious result with direct implications for the bounded rationality literature.

The primary limitation of the results is the coarseness of the card abstraction. The 100-bucket post-flop abstraction merges strategically distinct hands into the same information sets, which underestimates the value of information relative to the true Nash Equilibrium. The direction of the effects is robust, but specific magnitudes should be interpreted as approximations rather than exact equilibrium values.

Several extensions would strengthen these findings with additional time and resources. First, the current design always pits the behavioral archetype against a Nash opponent. A natural extension would simulate

matches between two behaviorally distorted players, as in real-world settings, perfectly rational players are rare and most exhibit some form of preference distortion. Second, the loss-averse nit archetype was excluded due to computational issues with the solver; a straightforward fix would involve either switching to floating-point regret storage or rescaling chip values, and the nit results would meaningfully complete the archetype comparison. Third, the current design has stack depths as discrete where a continuous treatment of stack depth would allow for more precise identification of exact effects. Fourth and most importantly, the card abstraction could be refined: increasing from 100 to 500 or 1000 post-flop buckets would reduce abstraction-induced bias, at the cost of substantially higher memory and compute requirements.

6 Conclusion

This paper set out to answer a simple yet underexplored question: how do capital constraints and behavioral distortions interact in competitive zero-sum settings, and can rational play substitute for capital when facing a less rational opponent? The results provide a nuanced answer. Stack asymmetry imposes a penalty on the short-stacked player that is non-monotonic. Moderate asymmetry is partially offset by commitment credibility effects, while extreme asymmetry compounds disproportionately. Behavioral distortions are exploitable by a rational opponent across all stack depths, and the two effects do not interact uniformly. Several extensions would strengthen these findings with additional time and resources. A natural next step would be to decompose the cost of suboptimal play into direct and indirect components — how much of the EV loss stems from the distorted player's own choices versus how much is attributable to the rational opponent actively exploiting the behavioral bias. A richer set of behavioral archetypes could also be explored, including distortions that are card-dependent rather than action-dependent. For instance, a player who overcalls when drawing to a straight or flush, a pattern well-documented in real poker hand histories. The loss-averse nit archetype, excluded here due to computational constraints, remains a theoretically motivated extension that the right implementation would make tractable. Many interesting behavioral questions also extend beyond the play of a single hand. The tools developed here are potentially capable of analyzing tournament structures, where survival across multiple hands introduces dynamic considerations that single-hand equilibrium analysis cannot capture, for example, stack management, risk of ruin, and the strategic value of chip preservation. These extensions are left for future work.

A Uniqueness of the Nash Equilibrium Payoff in Two-Player Zero-Sum Games

In a two-player zero-sum game, equilibrium strategies need not be unique, but the equilibrium payoff is unique. The reason follows from the zero-sum structure and the minimax theorem.

Zero-Sum Structure

Let player 1's payoff be $u_1(s_1, s_2)$ and player 2's payoff be

$$u_2(s_1, s_2) = -u_1(s_1, s_2).$$

Thus, player 1 maximizes u_1, while player 2 minimizes it.

Max-Min and Min-Max Problems

Player 1 solves

$$\max_{s_1} \min_{s_2} u_1(s_1, s_2),$$

and player 2 solves

$$\min_{s_2} \max_{s_1} u_1(s_1, s_2).$$

Minimax Theorem

By the minimax theorem,

$$\max_{s_1} \min_{s_2} u_1(s_1, s_2) = \min_{s_2} \max_{s_1} u_1(s_1, s_2) = v,$$

where v is the value of the game. This value is a single number and therefore unique.

Equilibrium Payoff

Let (s_1^*, s_2^*) be any Nash Equilibrium. Then:

- Since s_1^* is a best response to s_2^*,

$$u_1(s_1^*, s_2^*) \geq \max_{s_1} \min_{s_2} u_1(s_1, s_2) = v.$$

- Since s_2^* is a best response to s_1^*,

$$u_1(s_1^*, s_2^*) \leq \min_{s_2} \max_{s_1} u_1(s_1, s_2) = v.$$

Combining these inequalities yields $u_1(s_1^*, s_2^*) = v$.

Conclusion

Every Nash Equilibrium yields the same payoff v. While there may be multiple equilibrium strategies, the equilibrium payoff is unique because it coincides with the value of the game. Because a subgame perfect Nash Equilibrium is a Nash Equilibrium (SPNE refines the set of Nash Equilibria), it follows that all subgame perfect Nash Equilibria have payoff v as well.

B Glossary of Terms

Heads-Up A poker game played between exactly two players.

Limit Hold'em A variant of Texas Hold'em with fixed bet sizes.

Small Blind / Big Blind Forced bets posted before cards are dealt.

Small Bet / Big Bet The two fixed bet sizes used in limit hold'em.

N-Bet Cap A rule limiting the number of raises per betting round.

Stack Depth The number of chips a player holds, typically expressed in big blinds.

Position The order of action relative to the opponent.

Pot Odds The ratio of the current pot size to the cost of a call.

Showdown The end of a hand where cards are revealed.

Action Frequency The proportion of decisions at which a player takes a given action.

Street One of the four betting rounds: preflop, flop, turn, river.

Check Declining to bet when no bet is faced.

All-In Committing all remaining chips.

Game Tree A representation of all possible sequences of play.

MCCFR Monte Carlo Counterfactual Regret Minimization.

ACPC Format A standardized format for representing poker games.

Terminal Node A node at which the game ends.

Perfect Recall The assumption that players remember all past actions and observations.

References

Bowling, M., Burch, N., Johanson, M., & Tammelin, O. (2015). Heads-up limit hold'em poker is solved. *Science*, 347(6218), 145–149. https://doi.org/10.1126/science.1259433

Brown, N., & Sandholm, T. (2017). Libratus: The Superhuman AI for No-Limit Poker (Demonstration). https://www.ijcai.org/proceedings/2017/0772.pdf

Brown, N., & Sandholm, T. (2019). Superhuman AI for multiplayer poker. *Science*, 365(6456), 885–890. https://doi.org/10.1126/science.aay2400

Gibson, R. (2012). Efficient Monte Carlo Counterfactual Regret Minimization in Games with Many Player Actions. *Advances in Neural Information Processing Systems*, 25.

Gilpin, A., & Sandholm, T. (2007). Lossless abstraction of imperfect information games. *Journal of the ACM*, 54(5), 25. https://doi.org/10.1145/1284320.1284324

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.

Simon, H. A. (1955). A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1), 99–118.

Tammelin, O. (2014). Solving Large Imperfect Information Games Using CFR+. *arXiv*. https://arxiv.org/abs/1407.5042

Walasek, L., Mullett, T. L., & Stewart, N. (2024). A meta-analysis of loss aversion in risky contexts. *Journal of Economic Psychology*, 103, 102740. https://doi.org/10.1016/j.joep.2024.102740

Zinkevich, M., Johanson, M., Bowling, M., & Piccione, C. (2007). Regret Minimization in Games with Incomplete Information. https://poker.cs.ualberta.ca/publications/NIPS07-cfr.pdf